



Grant Agreement no. 101120075

HORIZON-WIDERA-2022-TALENTS-03-01
ERA Talents

D3.3

Protocols and standard operating procedures on practical recommendations covering the data driven innovative method validation into the clinical practice

Embryo image-analysis workflow

WP3 — Intersectoral mobility for R&I translation towards ethically responsible innovations

Document information	Details
Submission date	30 September 2026
Lead beneficiary	UM
Dissemination level	PU — Public
Authors	Pending
Reviewers	Pending
Version / status	0.2 - Draft
Deliverable reference	D3.3

Open-access web-based guidelines

<https://georgekazakis.github.io/nestor-sops/>

Summary

Deliverable D3.3 provides protocols, standard operating procedures and practical recommendations for validating data-driven innovative methods for use in clinical practice.

To this end, a practical workflow was established within NESTOR to study changes in the zona pellucida (ZP), inner cell mass (ICM) and trophectoderm (TE) during embryo development and hatching. This standard operating procedure combines automated segmentation, visual review guidelines and quantitative assessment of time-lapse images. It also includes filtering out frames with incomplete or incorrect segmentation and excluding embryos where segmentation quality is insufficient for analysis.

1. Introduction

1.1 Purpose

These guidelines draw on NESTOR research experience to support the evaluation of data-driven methods and their translation into clinical practice. They describe how to organise an analysis, review its outputs and document the evidence needed to assess its intended use.

Deliverable D3.3 forms part of WP3 and T3.1 of the [NESTOR project](#), with **UM** as the lead beneficiary. Its title is “Protocols and standard operating procedures on practical recommendations covering the data driven innovative method validation into the clinical practice”.

1.2 Scope

NESTOR brings together researchers in personalised reproductive medicine, assisted reproductive technologies (ART), non-invasive prenatal testing, uterine health and AI-based solutions for ART.

The guidelines presented here focus on embryo image analysis within the ART and AI research area. The [practical workflow](#) sets out the analysis and quality-review steps, while the [embryo image-analysis example](#) shows how these steps were applied to study embryo development and hatching.

1.3 Project context

These guidelines contribute to NESTOR T3.1, which addresses the translation of research experience into open-access protocols and standard operating procedures.

The embryo-analysis work presented in these guidelines was carried out through a collaboration between members of INF and UM at the hospital in Maastricht.

2. Practical workflow

Evaluation of a data-driven method involves a defined research question, a documented dataset and an identifiable model. The workflow below organises these elements into five stages, from the definition of the task to the interpretation of findings. Each stage describes its purpose, the information involved and its contribution to the overall analysis.

The workflow provides a general framework for organising method evaluation. The data, performance measures and criteria for an acceptable result depend on the task being studied. A worked application is presented in the [embryo image-analysis example](#), using the same five stages.

Organising the evaluation

Step	Description	Documentation
1. Task definition	Definition of the research question, intended use and expected outputs.	Scope of the analysis and responsibilities for review.
2. Data description	Description of data sources, reference information and the separation of training and evaluation data.	Dataset description, preparation methods and selection records.
3. Model identification	Identification of the model version, configuration and meaning of its outputs.	Model record and analysis settings.
4. Evaluation and visual review	Assessment of performance and review of outputs before further analysis.	Evaluation results, review decisions and exclusion records.
5. Documentation and interpretation	Presentation of findings in relation to the research question and the limits of the analysis.	Results, supporting observations and further evaluation needs.

2.1. Task definition

The research question establishes what the method is intended to identify, measure or predict. Its intended use provides the context for interpreting the outputs: the same measurement may have different requirements when used for exploratory research or to support a particular decision.

The scope describes the population or material being studied, the features of interest and the expected outputs. It also identifies responsibilities for reviewing those outputs and the role of human assessment within the analysis. This makes the relationship between automated processing and expert judgement clear from the outset.

Criteria for an acceptable result follow from this scope. They describe the level of accuracy, completeness or consistency needed for the task and the types of error that would make an output unsuitable. A documented scope gives data selection, model evaluation and interpretation a common basis.

2.2. Data description

The dataset description covers the source of the data, how they were collected, the material included and any preparation applied before analysis. Relevant preparation may include image resizing, intensity normalisation or the handling of missing observations. These details help explain the conditions under which the method is evaluated.

When an analysis involves patient data, restrictions on data transfer may require the work to be performed within the hospital's approved computing environment. The arrangements for data access and processing therefore form part of the analysis setup.

Where evaluation involves comparison with reference information, its origin and review process form part of the description. For image analysis, this may include annotations identifying the structures of interest. Uncertainty or inconsistency in the reference information affects the interpretation of differences between the model and the reference.

The separation of training and evaluation data accounts for related observations from the same source, such as repeated measurements from one participant or images from one sequence. Their allocation to each set affects how independently the method is being evaluated.

Dataset documentation also describes the criteria for inclusion and exclusion. Links between the original observations, processed inputs and retained results make it possible to understand how the analysed dataset was formed. This is particularly relevant when quality review removes part of the original data.

2.3. Model identification

Model identification covers the version used for analysis, its associated configuration and the preparation of its inputs. The model and its settings together define the method being evaluated; a change in either may alter the resulting outputs.

The description of the outputs explains what each label, value or region represents. In an image-segmentation method, for example, the association between output labels and anatomical structures determines how overlays and subsequent measurements are interpreted. Units and any processing applied after prediction are also relevant to understanding the results.

A model record connects the input preparation, model version and analysis settings with the resulting evaluation. This supports reproducibility and allows comparisons between analyses to account for changes in the method, rather than attributing every difference to the data.

2.4. Evaluation and visual review

Evaluation examines how well the outputs address the task defined in step 1. Where suitable reference information is available, comparison with that reference provides a basis for calculating performance measures. The calculation and level of analysis are part of the evaluation description, since a result calculated per image may answer a different question from one calculated per sequence or participant.

Visual review complements quantitative measures in image-based applications. It reveals the nature and distribution of errors, including incomplete outputs or incorrect identification of structures that may be obscured by an overall performance value. Review of individual results also helps determine whether they can support subsequent measurements.

Quality review addresses three related aspects:

1. The characteristics that distinguish usable outputs from incomplete or incorrect results.
2. The level at which a decision applies, such as an individual observation or a complete sequence.
3. The basis for retaining, reprocessing or excluding the affected data.

The review criteria and reasons for exclusion explain how the final analysis set differs from the original dataset. Numerical thresholds and qualitative judgements represent different bases for these decisions and are described accordingly. Evaluation of the original outputs and analysis of the retained data answer different questions; the latter describes the subset that passed review.

Performance measures describe the quality of the method's outputs. Measurements subsequently derived from those outputs address the research question. Keeping these two types of result distinct clarifies what the evaluation establishes and how output quality affects the findings.

2.5. Documentation and interpretation

Documentation brings together the research question, dataset, model configuration, evaluation results and review decisions. Their connection allows a finding to be traced to the data and processing steps from which it was derived.

Quantitative results and qualitative observations provide different forms of evidence. Their presentation identifies how each was obtained, the observations to which it applies and any exclusions affecting its interpretation. This makes the basis of each finding clear without treating a visual judgement as an automatically calculated measurement.

Interpretation relates the findings to the task defined in step 1. It considers the coverage of the dataset, the quality of the reference information, the errors observed and the effect of data selection. These factors explain the conditions under which the findings are supported and the questions that remain open.

The resulting account describes both the contribution of the method and any further evaluation needed for its intended use. The [embryo image-analysis example](#) shows how the five stages relate to a specific NESTOR research application.

3. Embryo image-analysis example

This example presents the NESTOR study of embryo development and hatching through the five stages of the [practical workflow](#). It brings together the research objective, the image-analysis approach and the visual review procedure used to select segmentation outputs for measurement.

The embryo-analysis work was carried out through a collaboration between members of INF and UM at the hospital in Maastricht.

3.1. Task definition

The research objective was to study changes in the zona pellucida (ZP), inner cell mass (ICM) and trophoctoderm (TE) during embryo development and hatching. Automated segmentation identified these structures in time-lapse images, supporting the assessment of size, shape, ZP thickness and image intensity over time.

Visual observations and quantitative measurements were used together to examine developmental changes. Hatching was identified visually, providing an observation separate from the measurements extracted by the analysis pipeline.

3.2. Data description

The analysis used time-lapse image sequences of embryo development and hatching. Each sequence contained successive frames associated with an embryo, allowing changes in the identified structures to be followed over time.

Images, segmentation overlays and videos provided the material for visual assessment. Review determined which frames, and in some cases which entire embryo sequences, were retained for analysis and metric extraction.

The measurements were organised by embryo and frame. This preserved the relationship between each quantitative record and its position within the corresponding image sequence.

3.3. Model identification

The analysis pipeline used a segmentation model to identify ZP, ICM and TE in individual time-lapse frames. Its outputs represented the three anatomical structures and supported visualisation of their boundaries and extraction of quantitative measurements.

Segmentation overlays and videos allowed the predicted structures to be inspected in relation to the source images. The outputs also supported the assessment of image intensity and structural changes during development.

The model's role in this application was to identify the structures used for analysis. Visual review determined whether the resulting segmentation was suitable for measurement.

3.4. Evaluation and visual review

The quality-review procedure used in this application was based on visual inspection of the segmentation results for each run. It was comprised of three stages:

1. Visual inspection of the predicted ZP, ICM and TE structures.
2. Exclusion of frames with incomplete or incorrect segmentation from analysis.
3. Exclusion of an embryo from analysis and metric extraction when too many of its frames were incorrectly segmented.

The embryo-level decision relied on qualitative visual judgement, without a fixed numerical cutoff. The resulting measurements therefore describe the frames and embryos retained after this review.

Images, overlays and videos were examined alongside quantitative measurements to assess changes over time. This describes the visual quality review applied in the study; numerical segmentation-performance results are not presented in this example.

3.5. Documentation and interpretation

Quantitative measurements were stored in one CSV file per embryo, with one row per time-lapse frame. This organisation supported assessment of changes in size, shape, ZP thickness and image intensity across each sequence.

Hatching was assessed visually and was not recorded in the measurement CSV files. The visual observations and the quantities extracted by the pipeline therefore provide distinct sources of information about development and hatching.

Interpretation concerns the retained frames and embryos because incomplete or incorrect segmentation led to exclusions. The procedure illustrates how automated segmentation, visual review and quantitative assessment were combined in a research application. Its suitability for embryo selection or treatment decisions has not been established by this workflow alone.

4. Web-based guidelines

The accompanying website presents the general workflow and the embryo image-analysis example as open-access guidelines. The two pages follow the same five stages, connecting the general description with its application in the NESTOR study.

[NESTOR Clinical Validation Guidelines](#)